

1. Covariance Analysis in Health Insurance

Given Data

Premium Increase (x)

claims (y)

3.2

4.0

5.5

4.8

15

18

25

20

Step 1: Calculate Mean

$$\bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4} = 4.375$$

$$\bar{y} = \frac{15 + 18 + 25 + 20}{4} = 19.5$$

Step 2: Covariance Table

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

$$\sum (x - \bar{x})(y - \bar{y}) = 12.25$$

Covariance

$$\text{Cov}(x, y) = \frac{12.25}{4} = 3.0625$$

Result

$$\text{Covariance} = 3.0625$$

conclusion

Since the covariance is positive, there is a positive relationship between premium increase and claim frequency.

2. Data Smoothing using Equal-Frequency Binning

Given Data

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126,
126, 126, 126, 130, 135, 135, 138, 138, 138, 140,
145, 150, 152, 160, 180

Step 1

Total values = 27

Divide into 3 equal-frequency bins (9 values each).

Bin 1

110, 112, 115, 115, 118, 120, 120, 122, 124

Mean = 117.3

Smoothed values:

117.3, 117.3, 117.3, 117.3, 117.3, 117.3, 117.3, 117.3, 117.3

Bin 2

124, 126, 126, 126, 126, 130, 135, 135, 138

$$\text{Mean} = 129.6$$

Smoothed values:

129.6 repeated 9 times

Bin 3

138, 138, 138, 140, 145, 150, 152, 160, 180

$$\text{Mean} = 149$$

Smoothed values:

149 repeated 9 times

Result

The data is successfully smoothed using Equal-Frequency Binning.

3, Equal-width Binning for Risk Categorization
Given Data

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2,
30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Step 1

$$\text{Minimum} = 18.5$$

$$\text{Maximum} = 39.5$$

$$\text{Range} = 21$$

$$\text{Number of bins} = 3$$

$$\text{Bin width} = 21 \div 3 = 7$$

Bins

Bin 1: 18.5 - 25.5

Bin 2: 25.5 - 32.5

Bin 3: 32.5 - 39.5

After Boundary Smoothing

Bin 1

18.5, 18.5, 25.5, 25.5

Bin 2

25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5,
32.5.

Bin 3

32.5, 32.5, 32.5, 39.5, 39.5

Result

BMI values are grouped into 3 equal-width bins using boundary smoothing.

4. Data Reduction Using Bin Medians

Step 1

Arrange the values in ascending order.

Step 2

Divide into 3 bins (9 values each).

Median values

Bin 1 Median = 148

Bin 2 Median = 165

Bin 3 Median = 190

Smoothed Data

Bin 1

148 repeated 9 times

Bin 2

165 repeated 9 times

Bin 3

190 repeated 9 times

Result

Data reduction is successfully performed using the Bin Median method.

5. Standardization of Patient Age Data given data

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Step 1: Mean

$$\bar{x} = 33$$

(a) Mean Absolute Deviation (MAD)

Answer

MAD =

(b) Z -

(Standard

Result

The fir
standard

$$MAD = \frac{\sum |x - \bar{x}|}{n} = \frac{88}{10} = 8.8$$

Answer

$$MAD = 8.8$$

(b) Z - Score

(Standard Deviation ≈ 10.76)

Age	Z-score
18	-139
22	-102
25	-0.74
42	0.84

Result

The first four observations are successfully standardized using the Z-score method.